

No. of Printed Pages : 4
Roll No.

188443

Level 4, 2nd Sem / DVOC (Ref. & Air Cond.)

Subject : Basics of Thermodynamics

Time : 2 Hrs.

M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

- Q.1 Which of the following is an example of a fire tube boiler?
- a) Babcock and Wilcox boiler
 - b) Lancashire boiler
 - c) Benson boiler
 - d) Loeffler boiler
- Q.2 The purpose of a condenser in a steam power plant is to :
- a) Increase pressure
 - b) Remove impurities
 - c) Reduce back pressure on turbine
 - d) Store steam
- Q.3 In which cycle does heat addition take place at constant pressure and constant volume?
- a) Otto cycle
 - b) Diesel cycle
 - c) Carnot cycle
 - d) Dual cycle

Q.4 Which law of thermodynamics defines temperature equality between systems?

- a) First law b) Second law
- c) Zeroth law d) Third law

Q.5 Entropy is a measure of:

- a) Irreversibility or disorder
- b) Temperature
- c) Energy
- d) Pressure

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 The main function of a steam turbine is to convert thermal energy into _____ energy,

Q.7 The two main types of steam condensers are jet condensers and _____ condensers.

Q.8 In a four-stroke cycle, the four operations are intake, compression, _____, and exhaust.

Q.9 The first law of thermodynamics for a non-cyclic process is expressed as $\Delta Q = \Delta U + \underline{\hspace{2cm}}$.

Q.10 A perpetual motion machine of the first kind violates the _____ law of thermodynamics.

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SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

Q.11 Explain the differences between fire-tube and water-tube boilers with neat diagrams.

Q.12 What is reciprocating steam engine? Give their main parts and their working principle.

Q.13 Differentiate between reciprocating and rotary compressors.

Q.14 Define steam condenser and explain its objective.

Q.15 Explain the terms: swept volume clearance volume.

Q.16 Define internal energy and enthalpy.

Q.17 Describe and explain quasi-static process.

Q.18 Explain perpetual motion machine of first order.

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Describe the four -stroke petrol engine cycle with the help of a neat diagram.

Q.20 What is a Carnot cycle? Derive the expression for efficiency of a carnot heat engine.

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